UNIT–II: Computer Applications

1. Classification of Digital Computers

A **digital computer** is an electronic machine that processes data in digital form, mainly using binary digits **0 and 1**. It accepts data as input, processes it according to instructions, stores it when required, and produces meaningful information as output.

Digital computers can be classified mainly on the basis of their **size, processing capacity, storage capacity, number of users, and purpose**.

Classification on the Basis of Size and Capacity:

1. Microcomputer

A microcomputer is a small computer in which a **microprocessor** is used as the Central Processing Unit (CPU). It is generally designed for individual use.

Examples include desktop computers, laptops and personal computers.

Features:

- Small in size
- Relatively low cost
- Easy to operate
- Suitable for individual users
- Used in homes, offices, schools and businesses

Business Uses: Preparing documents, maintaining accounts, billing, inventory management, e-mail, presentations and data analysis.

2. Minicomputer

A minicomputer is more powerful than a microcomputer and can support several users at the same time. It was traditionally used by medium-sized organizations and departments.

Features:

- Medium processing capacity
- Supports multiple users
- Greater storage than traditional microcomputers
- Suitable for departmental applications

Uses: Business data processing, manufacturing control, scientific applications and organizational databases.

3. Mainframe Computer

A mainframe is a large and powerful computer designed to process a very large amount of data and serve hundreds or thousands of users and transactions.

Features:

- Very high processing capacity
- Large storage capacity
- Supports many users simultaneously
- Highly reliable
- Suitable for large organizations

Uses: Banks, insurance companies, railways, government departments and large corporations use mainframe systems for large-scale transaction processing.

Example: Banks may use mainframe systems to process millions of customer transactions.

4. Supercomputer

A supercomputer is an extremely powerful computer designed to perform very complex calculations at very high speed.

Features:

- Extremely high processing speed
- Very large computational capacity
- Used for complex scientific calculations
- Expensive and sophisticated

Uses: Weather forecasting, climate research, space research, scientific simulations, defence research and other computationally intensive applications.

Classification on the Basis of Purpose

Digital computers may also be classified as:

General-Purpose Computers: These computers can perform many different types of tasks. For example, a PC can be used for accounting, word processing, internet access and preparing presentations.

Special-Purpose Computers: These are designed to perform a particular task or a limited set of tasks. Examples include computers embedded in ATMs, industrial control systems and specialized equipment.

2. Characteristics of Computers

A computer has several characteristics that make it useful in business and other fields.

1. Speed

A computer can perform millions or billions of operations in a very short period of time. Tasks that may take humans several hours can often be completed by computers within seconds.

2. Accuracy

Computers can produce highly accurate results when correct data and instructions are provided. Errors usually arise because of incorrect input, instructions or programs rather than because the computer becomes careless.

This is commonly expressed through the concept **GIGO – Garbage In, Garbage Out**.

3. Diligence

A computer does not become tired or bored. It can perform repetitive tasks continuously while maintaining the same level of speed and accuracy.

4. Storage Capacity

Computers can store a huge quantity of data and retrieve it quickly whenever required. Documents, customer records, accounts, images and other information can be stored digitally.

5. Automation

Once data and instructions are supplied, a computer can perform many operations automatically with little or no human intervention.

6. Versatility

A computer can perform different types of work. The same computer can be used for accounting, payroll, preparing documents, presentations, communication and data analysis.

7. Reliability

Computers can perform repeated operations consistently and reliably. Therefore, organizations depend on them for many routine business activities.

8. Multitasking

Modern computers can handle several tasks or applications at the same time. For example, a user may work on a document while a file is downloading and another application is running.

Important Exam Point

A computer has **no intelligence or feelings of its own**. It works according to the instructions given by the user or program.

3. Advantages of Computers

Computers provide many advantages to individuals and business organizations.

High Speed: Computers process large amounts of information quickly and save considerable time.

Accuracy: They provide accurate results when the input data and instructions are correct.

Large Storage: A large quantity of business information can be stored in digital form.

Automation: Routine operations such as payroll calculation and report generation can be automated.

Reduction in Repetitive Work: Computers efficiently perform repetitive tasks and reduce manual effort.

Better Communication: E-mail, messaging, video conferencing and other digital communication systems make business communication faster.

Better Decision-Making: Managers can use computer-generated reports and data analysis for planning and decision-making.

Cost Effectiveness: Although computers require an initial investment, automation can reduce operating costs and increase productivity over time.

Improved Customer Service: Businesses can maintain customer databases, process orders quickly and provide faster services.

Easy Data Retrieval: Stored information can be searched, accessed and updated quickly.

Exam Conclusion

Therefore, computers have become an essential part of modern business because they provide speed, accuracy, storage, automation and efficient information processing.

4. Computer Languages

A **computer language** is a formal language used to give instructions to a computer. A set of instructions written to perform a specific task is called a **program**.

Computer languages are commonly studied in the following categories.

1. Machine Language

Machine language is the **lowest-level programming language** and consists of binary digits, that is, **0 and 1**.

Example:

```
10110000 01100001
```

The computer's processor can execute machine instructions directly.

Advantages: It does not require translation into a lower-level form for execution and provides direct control over the hardware.

Disadvantages: It is difficult for humans to write, understand and debug. It is also machine-dependent.

2. Assembly Language

Assembly language uses **mnemonic codes or symbolic instructions** instead of writing instructions entirely in binary form.

Examples of mnemonics include:

`ADD`, `SUB`, `MOV`

Assembly language is easier for humans to understand than machine language.

A translator called an **Assembler** converts assembly-language instructions into machine code.

Advantages: Easier than machine language and provides relatively direct hardware control.

Disadvantages: It remains machine-dependent and requires detailed knowledge of computer architecture.

3. High-Level Language

A high-level language uses instructions and syntax that are much easier for humans to read and write than machine or assembly language.

Examples include **C, C++, Java, Python, BASIC and COBOL**.

Advantages:

- Easier to learn and write
- Easier to debug and maintain
- Programs can generally be adapted across different computer systems more easily than low-level programs
- Suitable for developing large applications

A high-level program must be translated into machine code before execution.

Language Translators

Assembler

An assembler converts an **assembly-language program into machine code**.

Compiler

A compiler generally translates a **complete high-level program into machine/object code** before execution and reports detected errors.

Interpreter

An interpreter translates and executes a program **instruction by instruction** rather than translating the entire program in one step.

Important Difference

Language	Form	Translator	Main Feature
Machine Language	Binary 0 and 1	Not normally required	Directly executable by processor
Assembly Language	Mnemonics	Assembler	Easier than machine language
High-Level Language	Human-readable syntax	Compiler/Interpreter	Easier programming and maintenance

Exam Conclusion

Computer languages provide a means of communication between programmers and computer systems. Development from machine language to high-level languages has made programming easier, faster and more accessible.

5. Personal Computers in Business

A **Personal Computer (PC)** is a microcomputer primarily designed for use by an individual. PCs are widely used in business because they are relatively affordable, versatile and capable of performing many office and managerial functions.

Common forms include **desktop computers and laptops**.

Applications of Personal Computers in Business

1. Word Processing

PCs are used for preparing letters, reports, notices, business documents and other written communication through word-processing software.

2. Accounting

Businesses use computers for recording financial transactions, maintaining accounts, preparing financial statements and supporting taxation-related work.

3. Payroll Management

Computers help calculate salaries, allowances, deductions and other payroll information of employees.

4. Inventory Management

PCs can maintain information about stock received, stock sold and stock available. This helps businesses manage inventory more efficiently.

5. Billing and Invoicing

Businesses use computers to prepare invoices, bills, receipts and sales records quickly and accurately.

6. Database Management

Customer details, employee records, supplier information and product details can be stored and managed using database systems.

7. Spreadsheet and Data Analysis

Spreadsheet software can be used for calculations, budgeting, forecasting, preparing tables and analysing business data.

8. Business Communication

PCs facilitate communication through e-mail, messaging, video meetings and other internet-based communication services.

9. Presentation

Managers and employees can prepare presentations containing text, tables, charts, graphics and other information for meetings, training and sales activities.

10. Internet and E-Commerce

Businesses use PCs for online banking, e-commerce, digital marketing, customer support, purchasing and gathering market information.

11. Management Information

Personal computers help managers prepare reports and analyse organizational data, thereby supporting planning, control and decision-making.

Advantages of PCs in Business

Personal computers help businesses to **save time, reduce paperwork, increase accuracy, improve communication, maintain records efficiently, analyse data and increase employee productivity**.

6. Introduction to Operating System

An **Operating System (OS)** is system software that acts as an **interface between the user and computer hardware**. It controls and coordinates the various resources of a computer and provides an environment in which application programs can run.

In simple words, an operating system manages the overall working of a computer.

Examples: Microsoft Windows, Linux, UNIX, macOS, Android and iOS.

Main Functions of an Operating System

1. Process Management: The operating system manages different programs and processes running on the computer. It decides how CPU time is allocated among them.

2. Memory Management: It manages the main memory (RAM) and allocates memory to different programs according to their requirements.

3. File Management: It helps in creating, storing, organizing, renaming, copying and deleting files and folders.

4. Device Management: The OS controls input and output devices such as keyboard, mouse, printer, scanner and storage devices.

5. User Interface: It provides a means through which the user can interact with the computer. The interface may be command-based or graphical.

6. Security: It provides security mechanisms such as user accounts, passwords, permissions and access control.

7. Resource Management: It manages computer resources such as CPU, memory, storage and peripheral devices.

8. Error Handling: The operating system detects many hardware and software errors and takes appropriate action or informs the user.

Conclusion

Thus, an operating system is essential system software because it controls computer resources and provides a link between users, applications and hardware.

7. Various Types of Operating Systems

Operating systems can be classified according to their working method and the number of users or tasks they support.

1. Single-User Operating System

A single-user operating system is primarily designed for one user at a time.

Example: MS-DOS was commonly used as a single-user operating system.

2. Multi-User Operating System

A multi-user operating system allows multiple users to access the resources of a computer system.

Example: UNIX and Linux systems can support multi-user environments.

3. Single-Tasking Operating System

A single-tasking operating system allows a user to perform essentially one main task at a time.

Example: Traditional MS-DOS environments.

4. Multitasking Operating System

A multitasking operating system allows several programs or tasks to operate apparently at the same time by efficiently sharing processor resources.

For example, a user may work on a document while playing audio and downloading a file.

Examples: Windows, macOS and Linux.

5. Multiprocessing Operating System

A multiprocessing operating system is capable of using two or more processors/processing cores to execute tasks.

It improves performance and allows efficient processing of multiple operations.

6. Real-Time Operating System (RTOS)

A real-time operating system processes data and responds within a defined time limit. It is used where timely and predictable responses are important.

Uses: Industrial control systems, robotics, medical equipment and embedded systems.

7. Network Operating System

A network operating system provides services for managing computers and other resources connected through a network.

It may provide facilities such as file sharing, printer sharing, centralized user management and network security.

Examples: Windows Server and various Linux server distributions.

8. Windows Operating System

Microsoft Windows is a widely used family of operating systems developed by Microsoft. It provides a **Graphical User Interface (GUI)** that allows users to interact with the computer through windows, icons, menus, buttons and a pointer.

Windows is widely used on personal computers in homes, educational institutions and business organizations.

Important Features of Windows

1. Graphical User Interface

Windows provides a graphical environment with icons, menus, windows, buttons and other visual elements, making the computer easier to operate.

2. Multitasking

Users can run multiple applications during the same session. For example, a user can work in a spreadsheet while a web browser and media application are also running.

3. File and Folder Management

Windows provides tools such as **File Explorer** to create, copy, move, rename, search and delete files and folders.

4. Hardware Support

Windows supports a wide variety of hardware devices such as printers, keyboards, mice, monitors, storage devices and scanners through appropriate drivers.

5. Networking

It provides facilities for internet access, network connectivity and sharing resources such as files and printers.

6. Security

Modern versions of Windows provide user accounts, passwords, access permissions, firewall facilities, security updates and other security mechanisms.

7. Multimedia Support

Windows supports audio, video, graphics, animation and various multimedia applications.

8. Search Facility

Users can search for files, folders, applications and settings.

9. Software Compatibility

A large variety of business, educational and general-purpose applications are available for Windows.

10. User-Friendly Environment

Its graphical interface makes routine computer operations relatively easy to learn and perform.

9. Advantages of Windows Operating System

Easy to Use: Its GUI makes it easier for users to interact with computers.

Multitasking: Several applications can be used during the same session.

Wide Software Availability: A large number of applications are developed for Windows.

Hardware Compatibility: Windows supports a broad range of computer hardware and peripheral devices.

Networking Support: It provides facilities for internet connectivity, local networks and resource sharing.

File Management: Files and folders can be easily created, organized, copied, moved, searched and deleted.

Multimedia Support: It provides extensive support for sound, video, images and other multimedia applications.

Useful for Business: Windows-based computers can be used for accounting, office applications, database management, communication, presentations and other business operations.

Exam Conclusion

Windows is a popular operating system because of its graphical interface, ease of use, multitasking capability, broad software ecosystem and hardware support.

10. CUI – Character User Interface

CUI stands for Character User Interface. It is also commonly discussed as a command-based or command-line interface because users interact with the computer mainly by **typing commands using the keyboard**.

Instead of selecting graphical icons and menus, users must enter appropriate commands.

Example: Traditional MS-DOS command environment.

Typical commands include:

`DIR` – displays files and directories
`CD` – changes directory
`COPY` – copies files
`DEL` – deletes files

Features of CUI

- Mainly text/character-based.
- Commands are entered through the keyboard.
- Users need to know the correct commands and syntax.
- It can consume relatively few graphical system resources.
- It can provide fast and precise control for experienced users.

Advantages of CUI

- Requires fewer graphical resources.
- Can be fast for experienced users.
- Useful for command-based administration and automation.
- Provides direct control through commands.

Limitations of CUI

- Commands must be learned or remembered.
- Less convenient for beginners.
- Incorrect commands may lead to errors.
- It is less visually intuitive than a graphical interface.

11. GUI – Graphical User Interface

GUI stands for Graphical User Interface. It allows users to interact with a computer using graphical elements such as **windows, icons, menus, buttons and a pointer**.

Instead of memorizing commands, users can select options using a mouse, touchpad, touchscreen or keyboard.

Examples: Microsoft Windows and macOS provide graphical user interfaces; many Linux desktop environments also provide GUIs.

Features of GUI

- Uses icons, menus and windows.
- Supports pointing and clicking.
- Easy to learn for most users.
- Provides visual feedback.
- Supports multiple windows and applications.
- Makes file and application management convenient.

Advantages of GUI

- User-friendly.
- Easy for beginners.
- Commands usually do not need to be memorized.
- Provides attractive visual representation.
- Makes multitasking and navigation easier.

Limitations of GUI

A GUI generally requires more memory, processing and graphics resources than a basic text interface. Some advanced operations may also be faster to perform through command-line tools for experienced users.

12. Difference Between CUI and GUI

Basis	CUI	GUI
Full Form	Character User Interface	Graphical User Interface
Interaction	Mainly typed commands	Icons, menus, windows, buttons
Input	Mainly keyboard	Mouse/touch + keyboard
Ease of Use	Requires command knowledge	Generally easier for beginners
Appearance	Mainly text based	Graphical and visual
Resource Requirement	Generally lower	Generally higher
Command Knowledge	Usually necessary	Much less necessary
Example	MS-DOS command environment	Microsoft Windows

13. System Software

System software is a collection of programs designed to control and manage computer hardware and provide a platform for application software.

It works mainly in the background and helps the computer system function efficiently.

Examples of System Software

Operating systems, device drivers, language translators and utility programs are major examples of system software.

14. Types of System Software

1. Operating System

An operating system manages computer hardware, software resources, files, memory, processes and input/output devices.

Examples: Windows, Linux, UNIX and macOS.

2. Device Drivers

A **device driver** is system software that enables the operating system to communicate properly with a particular hardware device.

Examples include printer drivers, graphics/display drivers and audio drivers.

Example: When a printer is connected to a computer, the appropriate printer driver helps the operating system communicate with and control the printer.

3. Language Translators

Language translators convert computer programs from one form into another form that the computer can execute.

Three important translators are:

Assembler: Converts assembly language into machine code.

Compiler: Translates a high-level program into machine/object code, generally as a complete program or compilation unit.

Interpreter: Translates and executes instructions progressively during program execution.

4. Utility Software

Utility software consists of programs designed to **maintain, protect, manage or optimize a computer system**.

Examples include:

- Antivirus and anti-malware tools
- Backup utilities
- Disk management tools
- File compression tools
- System diagnostic tools
- File-management utilities

Functions of Utility Software

Utility software helps improve system maintenance, protect data, manage storage, create backups and perform diagnostic or housekeeping operations.

5. Firmware

Firmware is software stored in non-volatile memory that provides low-level control or instructions for hardware devices.

For example, computers use firmware such as **BIOS/UEFI** during the startup process.

System Software vs Application Software

This distinction is frequently useful in examinations.

Basis	System Software	Application Software
Purpose	Manages computer system and resources	Performs user-specific tasks
Interaction	Often operates in the background	Used directly for user tasks

Dependency	Provides platform/services for applications	Generally, depends on system software
Examples	OS, drivers, utilities	Word processor, spreadsheet, browser
Main User	Computer system + users/applications	Primarily end users